

Clinical Trend Review — Hourly HR & RR

Data window: May 1, 2024 through June 30, 2024 (hourly min / max / average).

Important note: Trend review only; not a diagnosis. Interpret with age, meds, baseline rhythm, symptoms, and clinical context.

Data completeness: 1217 / 1464 hours (83.1%). Missing: 247 hours. Low-confidence (cnt<30): 29.

Largest data gaps (possible sensor downtime / out-of-range / connectivity):

Last recorded	Next recorded	Gap size
2024-06-29 19:00	2024-06-30 14:00	18 missing hr(s)
2024-06-24 11:00	2024-06-24 23:00	11 missing hr(s)
2024-05-26 20:00	2024-05-27 07:00	10 missing hr(s)
2024-06-21 21:00	2024-06-22 07:00	9 missing hr(s)
2024-05-22 21:00	2024-05-23 07:00	9 missing hr(s)
2024-05-25 21:00	2024-05-26 07:00	9 missing hr(s)
2024-06-03 21:00	2024-06-04 07:00	9 missing hr(s)
2024-06-01 21:00	2024-06-02 07:00	9 missing hr(s)

Summary statistics:

Metric	Mean	Median	P5	P95	Min	Max	SD
HR avg (bpm)	62.1	60.6	53.4	80.3	41.3	103.6	8.5
HR min (bpm)	55.6	54.6	48.2	65.8	40.1	99.5	7.4
HR max (bpm)	69.1	67.2	58.6	92.0	44.6	108.8	9.4
RR avg (brpm)	19.1	19.5	11.7	22.3	9.7	29.1	2.8
RR min (brpm)	15.2	15.5	9.6	18.3	8.4	21.3	2.3
RR max (brpm)	24.2	23.9	16.9	31.9	11.2	49.2	4.3

Threshold flags (adult-oriented reference points; interpret per patient baseline):

Flag	Count (hours)
Very high RR spikes (max RR >30)	78
Low confidence hour (cnt <30)	29
Severe bradycardia (min HR <45)	26
Bradycardia (avg HR <50)	20
Tachypnea (avg RR >24)	14
Tachycardia (avg HR >100)	7
Bradypnea (avg RR <10)	2

Notable episodes (contiguous hours):

Category	Start	End	Duration (hr)
Bradycardia (avg HR <50)	2024-06-23 22:00	2024-06-24 05:00	8
Bradycardia (avg HR <50)	2024-06-24 07:00	2024-06-24 11:00	5

Bradycardia (avg HR <50)	2024-06-25 02:00	2024-06-25 05:00	4
Bradycardia (avg HR <50)	2024-06-03 12:00	2024-06-03 12:00	1
Severe bradycardia (min HR <45)	2024-06-23 21:00	2024-06-24 05:00	9
Severe bradycardia (min HR <45)	2024-06-25 01:00	2024-06-25 05:00	5
Severe bradycardia (min HR <45)	2024-06-24 07:00	2024-06-24 10:00	4
Severe bradycardia (min HR <45)	2024-05-04 16:00	2024-05-04 17:00	2
Tachycardia (avg HR >100)	2024-06-27 20:00	2024-06-28 02:00	7
Bradypnea (avg RR <10)	2024-06-26 01:00	2024-06-26 01:00	1
Bradypnea (avg RR <10)	2024-06-26 10:00	2024-06-26 10:00	1
Tachypnea (avg RR >24)	2024-06-24 00:00	2024-06-24 09:00	10
Tachypnea (avg RR >24)	2024-05-12 00:00	2024-05-12 00:00	1
Tachypnea (avg RR >24)	2024-06-21 17:00	2024-06-21 17:00	1
Tachypnea (avg RR >24)	2024-06-23 17:00	2024-06-23 17:00	1

Clinical interpretation (trend-focused):

- **Heart rate:** Overall mean HR averages are in the low 60s bpm with intermittent bradycardic periods (avg HR <50 and/or min HR <45). A prominent bradycardic cluster occurs around **June 23–24**. Depending on baseline rhythm and medications, this may be physiologic (sleep) or clinically relevant (sinus node dysfunction, AV block, medication excess).
- **Tachycardic period:** A short cluster of tachycardia (avg HR >100) occurs around **June 27 evening through June 28 early morning**. Consider pain, infection, dehydration, anemia, volume overload, atrial arrhythmia, anxiety, or activity; correlate with BP, temperature, and rhythm data if available.
- **Respiratory rate:** Mean RR is ~19 brpm, with a small number of hours with avg RR >24 (tachypnea). Many hours show max RR >30, which can reflect brief bursts of rapid breathing or motion artifact; the avg RR is typically more reliable than max RR for trend assessment.
- **Data quality:** Overnight gaps suggest out of range/offline periods; interpret “episodes” cautiously when cnt is low or data are discontinuous.

Figure 1. HR trend with min/max lines and rolling means.

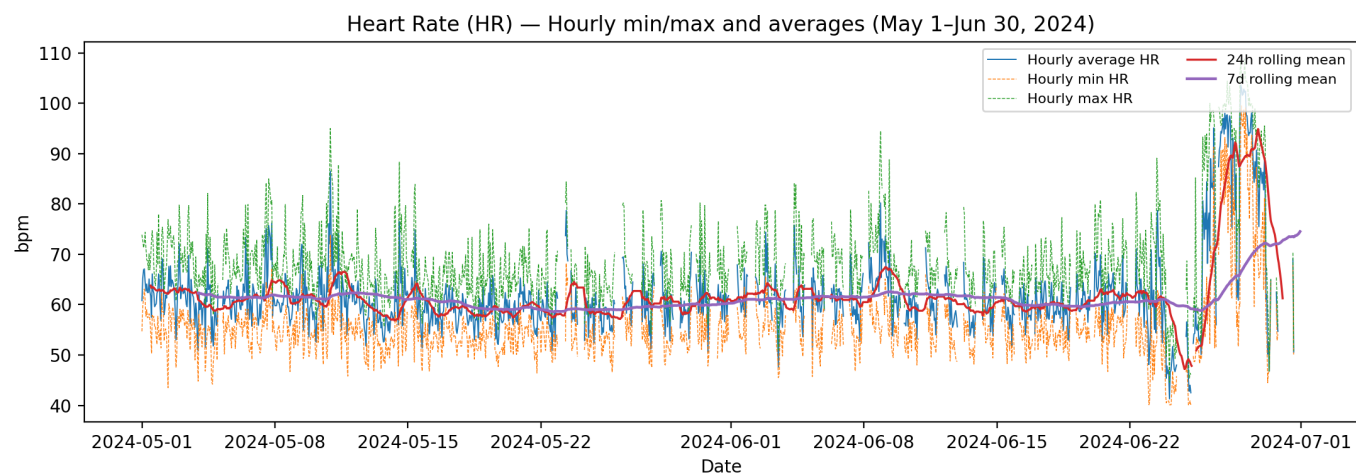


Figure 2. RR trend with min/max lines and rolling means.

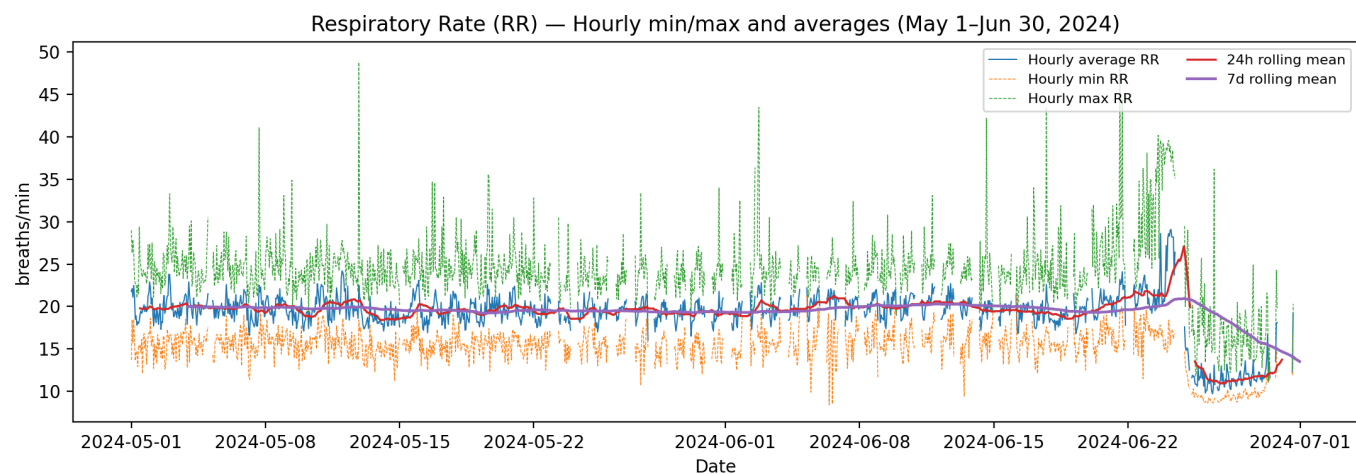


Figure 3. Mean diurnal pattern (hour-of-day).

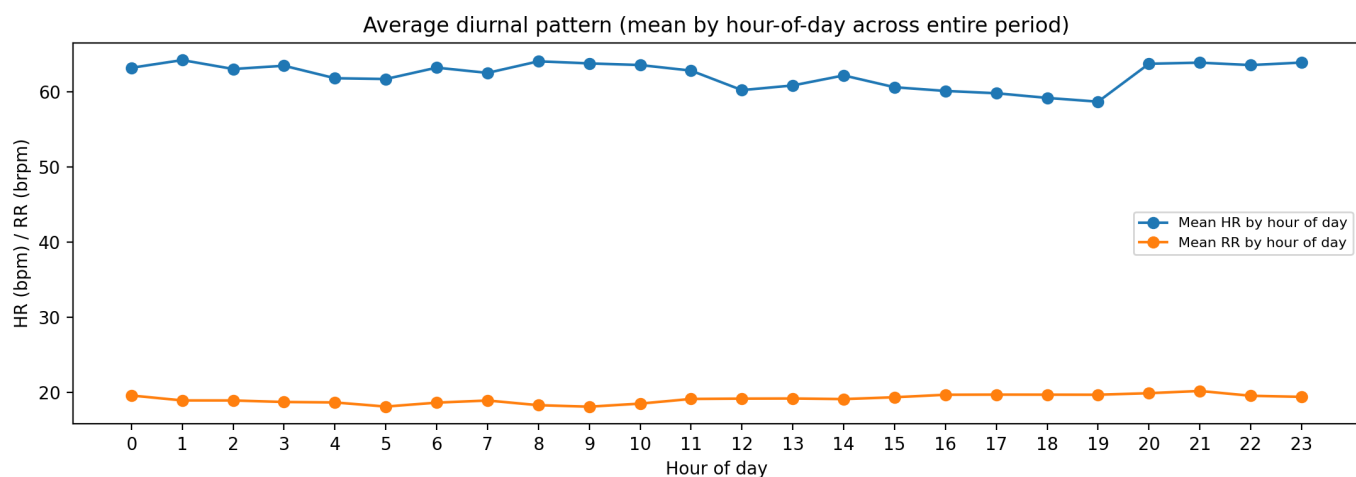
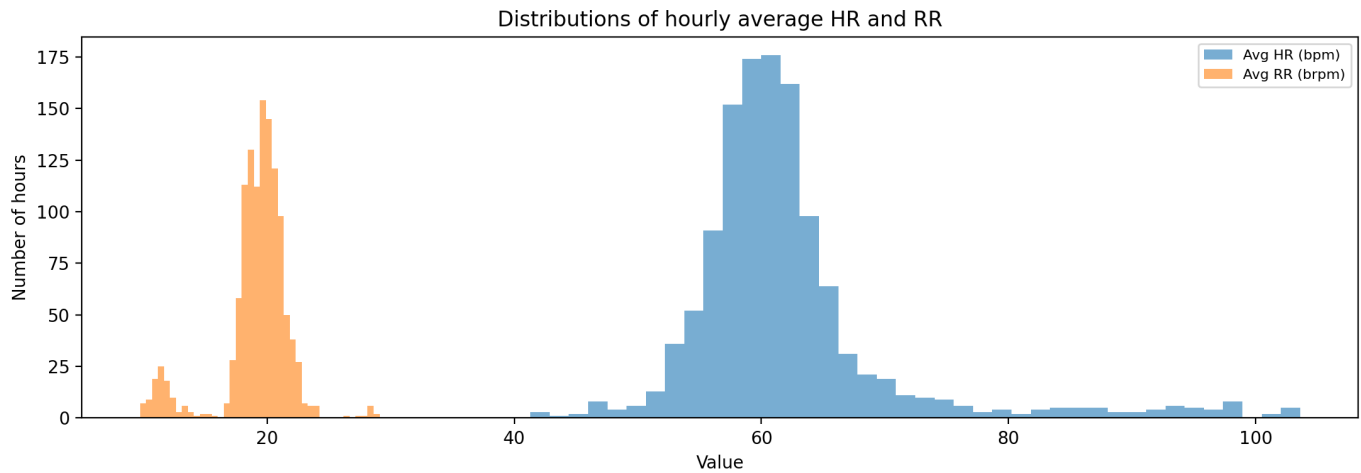


Figure 4. Distributions of hourly averages.



Suggested clinical follow-ups (for clinician review):

- 1) Review bradycardic windows (esp. June 23–24): assess symptoms, BP, medication timing/dosing; consider ECG review for pauses/AV block if available.
- 2) Review tachycardic window (June 27–28): correlate with symptoms and systemic drivers (infection, pain, volume status, anemia) and consider rhythm assessment during future events.
- 3) If RR elevations co-occur with HR rise or clinical symptoms, evaluate for respiratory/cardiac decompensation (SpO₂, lung exam, weight/fluid balance).
- 4) Improve continuity (reduce gaps / low-count hours) to increase actionable clinical signal.